

PhonePe Transaction Analysis

1. Project Overview

This project analyses transaction data to understand transaction volume, transaction value, payment trends, customer behavior and transaction success/failure patterns.

The Power BI dashboard converts raw transaction data into interactive visual insights that can support business decision-making.

2. Tools Used

- **Power BI** – Dashboard and visualization
- **Power Query** – Data cleaning and transformation
- **DAX** – KPI and calculated measures
- **Excel** – Data source

3. Dataset Structure

- **Table All_Transactions**
 - Rows : 101k
 - Columns : 8 (Transaction_ID, Amount, User_ID, Service, Service_Type, Payment_Status, Reason, Date)
- **Table All_Users**
 - Rows : 108k
 - Columns : 4 (User_ID, Name, Age, Join_Date)

4. Data Cleaning in Power Query

Before creating visuals, perform these checks:

- Check Column Headers
- Check Data profiling
- Check Data profiling
- Remove duplicates
- Handle blanks
- Replace values to standardize Data
- Created **Date Table** using DAX
- Created **Measures** in a table using DAX
- Added **Age Segment** column using **conditional column** for customer segmentation by age.

5. Data Modeling –

Create relationship between tables--

Mapped User_ID of All_Users table with User_ID of All_Transactions table as one (1) to many (*) relationship because one user can do multiple transactions

Mapped Date of Date Table with Date of All_Transactions table as one (1) to many (*) relationship because one user can have multiple transactions on one date.

6. Key KPIs

1. Total Transaction

Total Transaction = count(All_Transactions[Transaction_ID])

2. Total Transaction Value

Total transaction Values = sum(All_Transactions[Amount])

3. Successful Transaction

Successful Transaction = CALCULATE([Total Transaction],All_Transactions[Payment_Status]="Successful")

4. Success Rate

Success Rate = divide([Successful Transaction],[Total Transaction],0)

5. Total Users

Total Users = DISTINCTCOUNT(All_Users[User_ID])

6. Total Trans MOM%

Total Trans MOM% = divide([Total Transaction]-[Total trans PM],[Total trans PM],0)

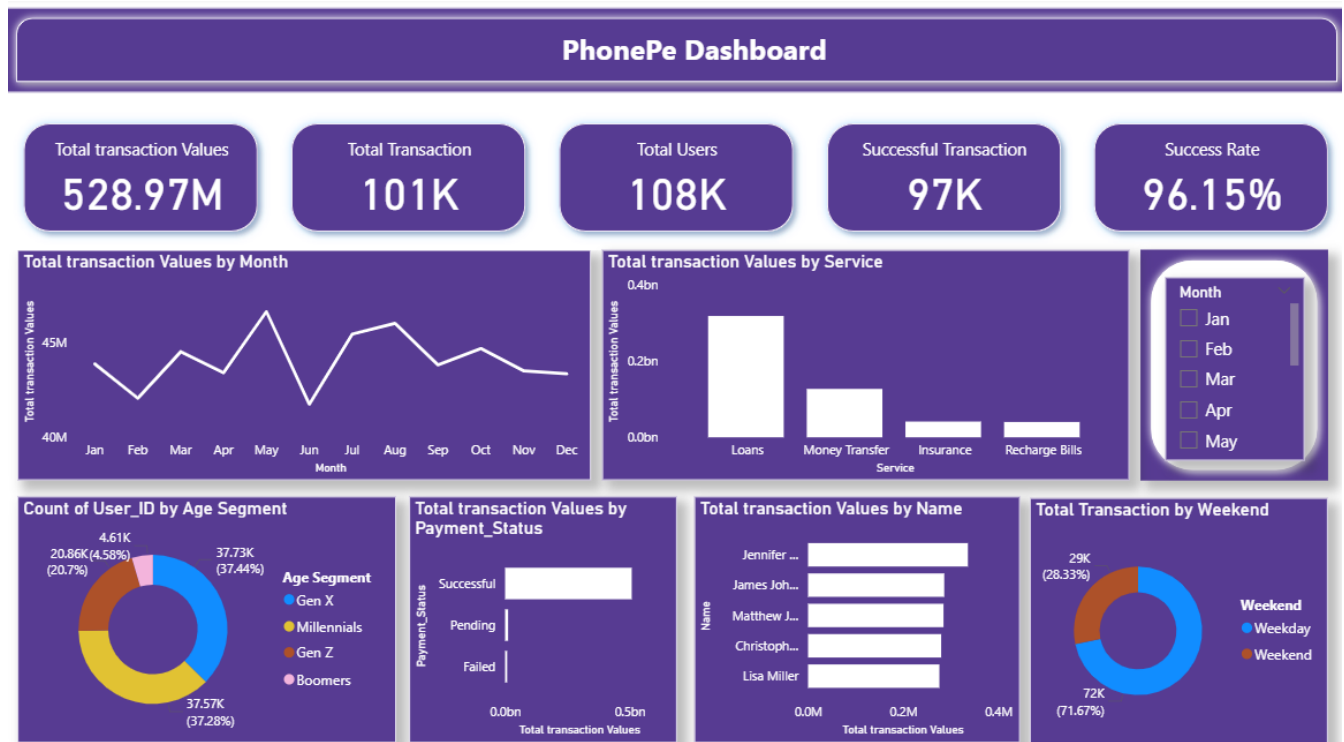
7. Trans Value MOM%

Trans Value MOM% = divide([Total transaction Values]-[Trans Value PM],[Trans Value PM],0)

7. Power BI Dashboard Design

Created 2 **report pages** in Dashboard for better understanding:

Page 1 — Executive Overview



KPI Cards

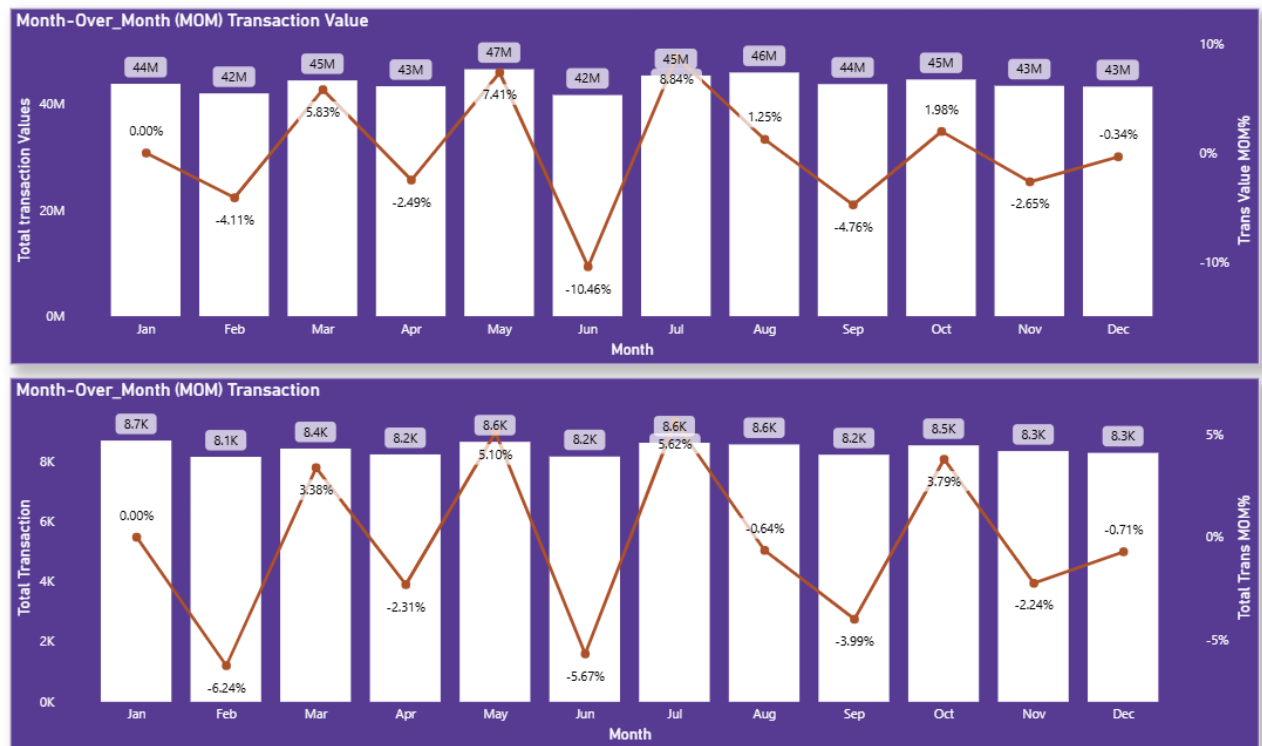
- Total Transaction Value
- Total Transaction
- Total Users
- Successful Transaction
- Success Rate

Visuals

- **Total Transaction Value by Month:** Line Chart for each month transaction value.
- **Total Transaction Value by Service:** Column Chart to know the service which has high transaction value and which has low.
- **Total Users by Age:** Doughnut Chart to find which age group of customers having most transaction value.
- **Total Transaction Value by Payment_Status:** Bar Chart to find the status of payments whether they are Successful, Pending or Fail.
- **Top 5 Users by Transaction Value:** Bar Chart to find top 5 users who have maximum transaction value.
- **Weekend Transaction Status:** Doughnut Chart to compare the no of transactions on weekday and weekend.

9. Page 2 — MOM Analysis

This page focuses on Month-Over-Month performance.



- **Month_Over_Month Transaction Value:** Line And Clustered Column Chart to measure the change in performance between one month and the immediately preceding month in terms of value of transactions.
- **Month_Over_Month Transaction:** Line And Clustered Column Chart to measure the change in performance between one month and the immediately preceding month in terms of number of transactions.

12. Important Business Insights

- **Transaction Trend:** Transaction volume shows an increasing/decreasing trend over the analysed period, with certain months recording significantly higher activity.
- **High Value Customer:** Find top 5 high value customers. Focus on these customers to maintain relationship in future.
- **Customer Segmentation:** Analysed which age group of customers are performing the high no of transactions.
- **Transaction Type:** Certain transaction types account for a larger share of total transaction activity, indicating stronger customer adoption of those services.

13. Business Recommendations

Based on the findings, PhonePe could:

- **Promote underutilized transaction services** through targeted campaigns.
- **Monitor transaction trends** to prepare infrastructure for peak periods.
- **Improve customer support** for failed or unsuccessful transactions.
- **Analyse high-value transaction segments** to understand customer behavior.